

KPI DEFINITION DOCUMENT

U.S. Medicaid Drug Utilization & Cost Intelligence

A 10-Year State-Level Analysis (2016–2025)

Document Title	KPI Definition Document — Medicaid Drug Spend Intelligence & Cost Optimization
Prepared By	Bhumi Nagar — Business Analyst
Date	April 2026 Version 3.0
Primary Stakeholder	Medicaid Managed Care Organization (MCO) & Payer Analytics Team
Total KPIs Defined	10 KPIs across 5 analytical dimensions

1. PURPOSE & OVERVIEW

This KPI Definition Document establishes the standardized definitions, business context, calculation logic, and interpretation guidance for all key performance indicators used in the Medicaid Drug Spend Intelligence & Cost Optimization project.

The purpose of this document is to:

- Standardize KPI definitions across all analytical deliverables
- Ensure consistency between SQL views and Power BI dashboard
- Document business calculations and analytical logic
- Improve stakeholder understanding of all metrics
- Support analytical transparency across all dashboard pages

2. KPI CATEGORIES

The 15 project KPIs are organized into 6 analytical categories, each serving a distinct business purpose for the MCO payer analytics team:

Category	Purpose	KPIs Included
Financial KPIs	Measure pharmaceutical spending and reimbursement trends	Total Spend, Medicaid Spend, Non-Medicaid Spend
Utilization KPIs	Measure prescription and unit activity volumes	Total Prescriptions, Total Units
Efficiency KPIs	Measure cost efficiency and per-unit pricing behavior	Avg Cost/Rx, Avg Cost/Unit, Avg Units/Rx
Dependency KPIs	Measure the Medicaid reimbursement dependency ratio	Medicaid Dependency Ratio
Concentration KPIs	Identify cost and utilization concentration patterns	Spend Share, Prescription Share, Cumulative Spend Share

Data Quality KPIs	Measure data suppression rates and reporting reliability	Suppressed Rows, Non-Suppressed Rows, Suppression Rate
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3. KPI SUMMARY INDEX

#	KPI Name	Category	Source Table	Dashboard Page	Type
1	Total Spend	Financial	kpi_yearly, kpi_state, kpi_drug	All Pages	SUM
2	Medicaid Spend	Financial	kpi_yearly, kpi_state	Executive, Trends	SUM
3	Non-Medicaid Spend	Financial	kpi_utilization_type	Cost Drivers	SUM
4	Total Prescriptions	Utilization	kpi_yearly, kpi_drug	All Pages	SUM
5	Total Units	Utilization	kpi_yearly, kpi_drug	Executive, Drug Insights	SUM
6	Avg Cost per Prescription	Efficiency	All KPI views	Executive, Cost Drivers	Calc
7	Avg Cost per Unit	Efficiency	All KPI views	Drug Insights	Calc
8	Avg Units per Prescription	Efficiency	All KPI views	Drug Insights	Calc
9	Medicaid Dependency Ratio	Dependency	All KPI views	Executive, Trends	Calc
10	Spend Share	Concentration	kpi_drug_share	Drug Insights	Calc
11	Prescription Share	Concentration	kpi_drug_share	Drug Insights	Calc
12	Cumulative Spend Share	Concentration	kpi_drug_pareto	Drug Insights — Pareto	Calc
13	Suppressed Rows	Data Quality	kpi_suppression_year/state	Data Quality	Count
14	Non-Suppressed Rows	Data Quality	kpi_suppression_year/state	Data Quality	Count
15	Suppression Rate	Data Quality	kpi_suppression_year/state	Data Quality	Calc

4. KPI DEFINITIONS — DETAILED

Category 1 — Financial KPIs

KPI 01: Total Spend | Financial

Business Context

The single most important indicator of Medicaid pharmaceutical cost burden. For the MCO payer analytics team, this is the primary metric for monitoring program cost trajectory and identifying whether spending is growing at a sustainable rate.

Formula	SUM(total_amount_reimbursed)
Unit	USD (\$) displayed in Billions / Trillions
Aggregation Type	Sum
Dimensions	By Year By State By Drug By Utilization Type (MCOU / FFSU)
SQL Source	analytics.kpi_yearly kpi_state kpi_drug kpi_utilization_type
Dashboard Page	Executive Overview (KPI Card = \$1.5T) All Analytical Pages
Data Note	Suppressed records excluded — spend set to NULL where suppression_used = TRUE

Interpretation:

Signal	Business Interpretation
Rising	Pharmaceutical spending is increasing — signals rising drug costs, increased utilization, or new high-cost drug approvals. Requires drill-down to identify which drugs or states are driving growth.
Falling	Spend is declining or stabilizing — may reflect formulary management, rebate programs, or reduced utilization.
Watch	Rapid acceleration after 2020 — may reflect COVID-era utilization changes or new specialty drug approvals. Cross-reference with prescription volume.

- Limitation: Does not distinguish between price increases and volume increases — use Avg Cost per Prescription for decomposition

KPI 02: Medicaid Spend | Financial

Business Context	Isolates the government-funded portion of total pharmaceutical reimbursements. When Medicaid Spend tracks Total Spend closely, it confirms high government dependency in drug financing.
Formula	SUM(medicaid_amount_reimbursed)
Unit	USD (\$)
Aggregation Type	Sum
Dimensions	By Year By State
SQL Source	analytics.kpi_yearly kpi_state
Dashboard Page	Executive Overview Spending Trends Cost Drivers
Data Note	Complement: Non-Medicaid Spend = Total Spend – Medicaid Spend

Interpretation:

Signal	Business Interpretation
Rising	Government is funding a larger share of drug costs — signals increasing public sector pharmaceutical burden.

Falling	Non-Medicaid sources absorbing more spending may indicate a shift to commercial payer coverage.
Watch	Compared to the Medicaid Dependency Ratio for context, absolute spend change may differ from share change.

- Limitation: Does not capture rebate adjustments — gross spend reported, not net after manufacturer rebates

KPI 03: Non-Medicaid Spend | Financial

Business Context	Measures the reimbursement amounts not directly attributed to Medicaid programs. Comparing Non-Medicaid Spend against Medicaid Spend reveals the true payer split and helps assess how much cost burden falls on non-government sources.
Formula	SUM(non_medicaid_amount_reimbursed)
Unit	USD (\$)
Aggregation Type	Sum
Dimensions	By Utilization Type By Year By State
SQL Source	analytics.kpi_utilization_type kpi_yearly kpi_state
Dashboard Page	Cost Drivers Utilization Type Analysis
Data Note	Low values expected — given 95.8% Medicaid Dependency Ratio, Non-Medicaid Spend is a small fraction of total

Interpretation:

Signal	Business Interpretation
Rising	Non-Medicaid spends rising — cost burden shifting toward non-government sources. May indicate increased co-pays or commercial payer coverage expansion.
Falling	Non-Medicaid spends are declining — the government is bearing a greater share of pharmaceutical costs.
Watch	Very low non-Medicaid spend relative to total — confirms structural Medicaid dependency. Policy risk if federal funding changes.

- Limitation: It may represent manufacturer rebates or patient co-pays — not necessarily independent payer coverage

Category 2 — Utilization KPIs

KPI 04: Total Prescriptions | Utilization

Business Context	Measures the volume of dispensing events across Medicaid programs. Used alongside Total Spend to determine whether cost growth is driven by more prescriptions (volume) or higher prices per prescription.
Formula	SUM(number_of_prescriptions)

Unit	Count — displayed in Billions
Aggregation Type	Sum
Dimensions	By Year By Drug
SQL Source	analytics.kpi_yearly kpi_drug
Dashboard Page	Executive Overview (KPI Card = 13bn) Spending Trends Drug Insights
Data Note	Suppressed records typically represent low-volume dispensing events

Interpretation:

Signal	Business Interpretation
Rising	Utilization growing — more beneficiaries receiving drugs or existing beneficiaries receiving more dispensing events.
Falling	Utilization declines may indicate formulary restrictions, access barriers, or disease management improvements.
Watch	Spend rising faster than prescriptions — confirms price-driven cost increases. Cross-reference with Avg Cost per Prescription.

- Limitation: Count does not capture drug strength, duration, or clinical appropriateness

KPI 05: Total Units | Utilization

Business Context	Measures the total number of medication units reimbursed across all Medicaid programs. Units represent the actual quantity of medication dispensed — pills, vials, ml, or other dose forms. Used alongside prescriptions to understand dosage intensity patterns.
Formula	SUM(units_reimbursed)
Unit	Count — displayed in Billions
Aggregation Type	Sum
Dimensions	By Year By Drug By State
SQL Source	analytics.kpi_yearly kpi_drug
Dashboard Page	Executive Overview (KPI Card = 833bn) Drug Insights
Data Note	Unit definition varies by drug — may represent tablets, vials, ml, or grams

Interpretation:

Signal	Business Interpretation
Rising	Total units are increasing, with more medication being dispensed. Could reflect increased disease prevalence, broader formulary coverage, or shift to higher-dose products.
Falling	Total units declined due to reduced dispensing volume. May reflect formulary restrictions or improved disease management.

Watch

Units growing faster than prescriptions — shift toward higher-unit supply per dispensing event (e.g., 90-day vs. 30-day supply).

- Limitation: Unit definition inconsistency across drugs limits direct drug-to-drug comparison

Category 3 — Efficiency KPIs

KPI 06: Avg Cost per Prescription | Efficiency

Business Context	The single most important pricing efficiency metric. Rising cost per prescription — even with flat or declining volume — is a direct signal of pharmaceutical price inflation and a primary driver of Medicaid spending growth.
Formula	$\text{SUM}(\text{total_amount_reimbursed}) / \text{SUM}(\text{number_of_prescriptions})$
Unit	USD per Prescription (\$)
Aggregation Type	Calculated Metric
Dimensions	By Year By State By Drug By Utilization Type
SQL Source	All KPI views
Dashboard Page	Executive Overview (KPI Card = \$307) Cost Drivers Drug Insights
Data Note	Calculated only for non-suppressed records

Interpretation:

Signal	Business Interpretation
Rising	Average dispensing cost rising — signals pharmaceutical price inflation. Compare against prescription volume to confirm price-driven vs. volume-driven cost growth.
Falling	Average dispensing cost stable or declining — may reflect formulary substitution with generic or drug patent expiration.
Watch	Large variation across states — may indicate different formulary structures or reimbursement rate negotiations. Investigate state outliers.

- Limitation: Average masks variation — a few extremely high-cost specialty drugs can significantly inflate the average
- Limitation: Gross reimbursement rates used — does not account for net cost after manufacturer rebates

KPI 07: Avg Cost per Unit | Efficiency

Business Context	Provides granular pricing efficiency at the dosage level. Critical for comparing drugs that come in different pack sizes, where prescription-level comparison would be misleading.
Formula	$\text{SUM}(\text{total_amount_reimbursed}) / \text{SUM}(\text{units_reimbursed})$
Unit	USD per Unit (\$)
Aggregation Type	Calculated Metric

Dimensions	By Year By State By Drug By Utilization Type
SQL Source	All KPI views
Dashboard Page	Drug Insights Cost Drivers
Data Note	Unit definition varies by drug — direct drug-to-drug comparison requires context

Interpretation:

Signal	Business Interpretation
Rising	Unit price rising — direct evidence of drug price inflation. Most concerning for high-spend specialty drugs, where the unit cost can be thousands of dollars.
Falling	Unit price declining — may indicate generic entry, biosimilar competition, or successful rebate negotiation.
Watch	High Cost per Unit + Low Units per Prescription = high-cost specialty drug. Cross-reference with total spending to assess program impact.

- Limitation: Unit definition inconsistency across drugs limits direct drug-to-drug comparison

KPI 08: Avg Units per Prescription | Efficiency

Business Context	Measures dosage intensity — how many units of a drug are dispensed per prescription event. Used with Cost per Unit to complete the cost decomposition picture.
Formula	$\text{SUM}(\text{units_reimbursed}) / \text{SUM}(\text{number_of_prescriptions})$
Unit	Units per Prescription
Aggregation Type	Calculated Metric
Dimensions	By Year By State By Drug By Utilization Type
SQL Source	All KPI views
Dashboard Page	Drug Insights
Data Note	High units per prescription may reflect 90-day supply dispensing — not necessarily higher dosage

Interpretation:

Signal	Business Interpretation
Rising	More units dispensed per prescription — may reflect shift to 90-day supply dispensing, higher dosage strength, or increased adherence programs.
Falling	Fewer units per prescription — may indicate 30-day supply dispensing, dose reduction, or therapeutic substitution.
Watch	Combine with Cost per Unit to determine if cost increases are unit-price driven or supply-quantity driven.

- Limitation: Unit definition varies — direct comparison across drug categories may be misleading

Category 4 — Dependency KPIs

KPI 09: Medicaid Dependency Ratio | *Dependency*

Business Context	Measures of the percentage of total pharmaceutical spend are funded directly by Medicaid programs. In this dataset, the ratio is approximately 95.8%, indicating extreme government dependency in drug financing.
Formula	$\text{SUM}(\text{medicaid_amount_reimbursed}) / \text{SUM}(\text{total_amount_reimbursed})$
Unit	Ratio (0.0–1.0) — displayed as Percentage (%)
Aggregation Type	Calculated Metric
Dimensions	By Year By State By Drug By Utilization Type
SQL Source	All KPI views (derived in staging.medicaid_enriched)
Dashboard Page	Executive Overview (KPI Card = 95.8%) Cost Drivers
Data Note	Values close to 1.0 indicate Medicaid funding nearly covers all costs

Interpretation:

Signal	Business Interpretation
Rising	Ratio increasing — government absorbing a greater share of pharmaceutical costs. Signals increased public sector fiscal exposure.
Falling	Ratio decreasing — cost burden shifting to non-government sources. May indicate commercial coverage expansion.
Watch	Ratio consistently above 90% structural dependency on government funding. Significant policy risk if federal Medicaid funding changes.

- Limitation: Non-Medicaid amount may represent manufacturer rebates or co-pays - not necessarily independent payer coverage

Category 5 — Concentration KPIs

KPI 10: Spend Share | *Concentration*

Business Context	Calculates each drug's percentage contribution to total Medicaid pharmaceutical expenditure. Foundational metric for Pareto analysis — directly informs formulary management priorities and rebate negotiation targets.
Formula	$\text{SUM}(\text{drug_total_spend}) / \text{SUM}(\text{all_drugs_total_spend})$
Unit	Percentage (%)
Aggregation Type	Calculated Metric
Dimensions	By Drug (ranked by spend)
SQL Source	analytics.kpi_drug_share

Dashboard Page	Drug Insights Pareto Chart
Data Note	Calculated only on non-suppressed records

Interpretation:

Signal	Business Interpretation
Rising	Single drug with high spend share (>5%) — high-cost specialty drug with significant formulary impact. Priority target for rebate negotiation.
Falling	Spend share declining for top drugs — may indicate patent expiration, biosimilar entry, or formulary substitution success.
Watch	Top 10 drugs accounting for >50% of spend — extreme concentration risk. Formulary disruption for any one drug could materially impact program costs.

- Limitation: Share is relative — a drug's share can increase without absolute spend increasing if other drugs' costs decline

KPI 11: Prescription Share | Concentration

Business Context	Measures each drug's percentage contribution to total prescription volume. When compared against Spend Share, it reveals whether a drug is high-cost due to price or due to high utilization volume — a critical distinction for formulary strategy.
Formula	$\text{SUM}(\text{drug_prescriptions}) / \text{SUM}(\text{all_drugs_prescriptions})$
Unit	Percentage (%)
Aggregation Type	Calculated Metric
Dimensions	By Drug
SQL Source	analytics.kpi_drug_share
Dashboard Page	Drug Insights Drug Share Analysis
Data Note	Compare against Spend Share to distinguish price-driven vs volume-driven cost concentration

Interpretation:

Signal	Business Interpretation
Rising	High Prescription Share — the drug is widely prescribed. If Spend Share is proportional, cost is volume-driven. If Spend Share is much higher, the drug has above-average pricing.
Falling	Low Prescription Share with high Spend Share — the drug is rarely prescribed but extremely expensive. Classic specialty/rare disease drug profile.
Watch	Drug with high Rx Share but low Spend Share — high-volume generic drug. Low unit cost, high utilization.

- Limitation: Does not capture prescription duration or supply quantity per dispensing event

KPI 12: Cumulative Spend Share / *Concentration*

Business Context	Running total of Spend Share when drugs are ranked from highest to lowest spend. Foundation of Pareto analysis — identifies exactly how many drugs account for 80% of total pharmaceutical spend. Key finding: ~20% of drugs drive ~80% of total Medicaid pharmaceutical spends.
Formula	Running SUM(spend_share) OVER (ORDER BY total_spend DESC)
Unit	Cumulative Percentage (%)
Aggregation Type	Window Function
Dimensions	By Drug (cumulative, ranked by spend DESC)
SQL Source	analytics.kpi_drug_pareto
Dashboard Page	Drug Insights — Pareto Chart
Data Note	WHERE cumulative_spend_share <= 0.80 isolates the drugs contributing to the first 80% of total spend

Interpretation:

Signal	Business Interpretation
Rising	80% threshold reached by fewer drugs — spend concentration increasing. Fewer drugs controlling more spend = higher formulary risk concentration.
Falling	80% threshold requires more drugs — spend distributing more evenly. Lower concentration risk.
Watch	Drugs before 80% threshold = Priority Tier 1 for formulary management, rebate negotiation, and utilization review.

- **KEY FINDING:** ~20% of drugs drive ~80% of total Medicaid pharmaceutical spend — confirmed by kpi_drug_pareto WHERE cumulative_spend_share <= 0.80
- **Limitation:** Historical concentration — does not predict future spend distribution if new specialty drugs enter market

Category 6 — Data Quality KPIs**KPI 13: Suppressed Rows** / *Data Quality*

Business Context	Counts the number of records suppressed by CMS — records withheld when data could identify specific prescribers or patients (typically low-volume records). Understanding suppression volume is essential for assessing analytical completeness.
Formula	COUNT(*) WHERE suppression_used = TRUE
Unit	Count (Number of Records)
Aggregation Type	Count
Dimensions	By Year By State By Utilization Type

SQL Source	analytics.kpi_suppression_year kpi_suppression_state kpi_suppression_utilization_type
Dashboard Page	Data Quality Page
Data Note	Suppressed records retained in dataset for counting but excluded from all spend/utilization calculations

Interpretation:

Signal	Business Interpretation
Rising	Suppressed rows increasing — more low-volume records being withheld. May reduce analytical completeness for certain drugs or states.
Falling	Suppressed rows declining — more records available for analysis. Increases confidence in KPI accuracy.
Watch	High suppression in specific states — state-level analysis for those states should be interpreted with caution and flagged in reporting.

- Limitation: Suppressed records represent real pharmaceutical activity — their exclusion means total spend may be slightly understated

KPI 14: Non-Suppressed Rows | Data Quality

Business Context	Counts the number of valid, non-suppressed records available for analytical use. This is the effective analytical population — all spend and utilization KPIs are calculated from non-suppressed records only.
Formula	COUNT(*) WHERE suppression_used = FALSE
Unit	Count (Number of Records)
Aggregation Type	Count
Dimensions	By Year By State By Utilization Type
SQL Source	analytics.kpi_suppression_year kpi_suppression_state
Dashboard Page	Data Quality Page
Data Note	This count represents the usable analytical records for all KPI calculations

Interpretation:

Signal	Business Interpretation
Rising	Non-suppressed rows increasing — more complete data available. Higher confidence in analytical accuracy.
Falling	Non-suppressed rows declining relative to total — suppression rate rising. Reduced analytical completeness.
Watch	Always review alongside the Suppression Rate to understand the proportion of usable vs. suppressed data.

- Limitation: Count alone is not sufficient — must be reviewed alongside the Suppression Rate for context

KPI 15: Suppression Rate | *Data Quality*

Business Context	Measures of the percentage of total records were suppressed. The suppression rate is the primary data quality indicator for this project — it determines how much of the analytical population is available and how reliable the spend and utilization of KPIs are.
Formula	COUNT(suppressed rows) / COUNT(total rows)
Unit	Percentage (%)
Aggregation Type	Calculated Metric
Dimensions	By Year By State By Utilization Type
SQL Source	analytics.kpi_suppression_year kpi_suppression_state kpi_suppression_utilization_type
Dashboard Page	Data Quality Page
Data Note	FFSU (Fee-for-Service) typically has higher suppression rates than MCOU (Managed Care) — affects comparative analysis reliability

Interpretation:

Signal	Business Interpretation
Rising	Suppression rate is increasing over time — CMS is applying stricter privacy standards. May reduce analytical completeness.
Falling	Suppression rate declines as more records are available. Increases confidence in spending and utilization KPI accuracy.
Watch	High suppression rate in specific states (>30%) - state-level analysis should be flagged with data quality caveats.

- Important: FFSU typically has higher suppression rates than MCOU — this affects MCOU vs FFSU comparative analysis reliability
- Limitation: Suppressed records represent real activity — total spend is likely slightly understated

5. KPI RELATIONSHIPS & COMBINED USAGE

The following KPI combinations deliver the most actionable business analysis:

Business Question	Primary KPI	Supporting KPI
Is cost growth price-driven or volume-driven?	Avg Cost per Prescription (trend)	Total Prescriptions (trend)
Which drugs are the top intervention targets?	Cumulative Spend Share (Pareto)	Spend Share + Total Spend
Is a drug expensive due to price or volume?	Spend Share vs Prescription Share	Avg Cost per Prescription
How is the government exposed to drug costs?	Medicaid Dependency Ratio	Medicaid Spend vs Non-Medicaid Spend

Is MCOU or FFSU more cost-efficient?	Avg Cost per Prescription by type	Total Spend + Prescriptions by type
Which states are high-cost outliers?	Total Spend by State	Avg Cost per Prescription by State
How reliable is the data for a given year/state?	Suppression Rate	Non-Suppressed Rows count

6. DASHBOARD KPI USAGE BY PAGE

The following table maps which KPIs appear on each Power BI dashboard page:

Dashboard Page	KPIs Used
Executive Overview	Total Spend Total Prescriptions Total Units Avg Cost per Prescription Medicaid Dependency Ratio
Spending Trends	Total Spend Total Prescriptions Medicaid Spend YoY growth patterns
State Performance	Total Spend by State Avg Cost per Prescription by State Medicaid Dependency Ratio by State
Drug Insights	Spend Share Prescription Share Cumulative Spend Share (Pareto) Avg Cost per Prescription Avg Cost per Unit Avg Units per Prescription
Cost Drivers	Avg Cost per Prescription (trend) Total Prescriptions (trend) Medicaid Spend Non-Medicaid Spend Medicaid Dependency Ratio
Data Quality	Suppressed Rows Non-Suppressed Rows Suppression Rate — by Year, State, and Utilization Type

7. KPI DESIGN PRINCIPLES

The KPI framework was designed based on the following principles:

Principle	Description
Business Relevance	Every KPI is tied to a specific business question for the MCO payer analytics team — no metrics exist for technical completeness alone
Executive Readability	KPIs are expressed in business-friendly units (USD, %, Count) and labeled clearly for non-technical decision-makers
Analytical Consistency	KPI calculations remain identical across all dashboard pages and SQL views — no metric is calculated differently in different contexts
Scalability	KPI views are built as modular SQL structures — supporting future addition of new years, states, drug categories, or analytical dimensions
Transparency	All data quality limitations, suppression handling, and exclusion decisions are documented — stakeholders are informed of what the data can and cannot show

8. FUTURE KPI ENHANCEMENTS

The following KPIs are identified as potential additions in future analytical phases:

- Year-over-Year (YoY) Growth Percentage — track spend and prescription growth rate annually

- Drug Category Contribution — aggregate spending and utilization by therapeutic category (requires external drug classification data)
- State Efficiency Index — composite metric ranking states by cost per prescription relative to the national average
- Forecasted Spend Metrics — predictive spend trajectory based on historical trend patterns
- Cost Variability Metrics — measure the standard deviation of cost per prescription across states for the same drug
- Rebate-Adjusted Spend — net spend after manufacturer rebate deductions (requires rebate data integration)
- Beneficiary-Level Metrics — per-member-per-month (PMPM) cost analysis (requires beneficiary enrollment data)

9. DOCUMENT SIGN-OFF

This KPI Definition Document has been prepared by Bhumi Nagar and establishes the standardized definitions, business context, and interpretation guidance for all 15 KPIs used in the Medicaid Drug Spend Intelligence project.

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